

Technology #175

Embed the enhanced physics-informed neural network model into the aircraft lithium-ion battery management system to continuously predict cell surface temperature during flight and trigger protective actions (e.g., load reduction, active cooling) when predicted temperature approaches runaway

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Embed the enhanced physics-informed neural network model into the aircraft lithium battery management system to continuously predict cell surface temperature during flight and trigger protective actions (e.g., load reduction, active cooling) when predicted temperature approaches runaway thresholds.

WHY DOES IT MATTER?

Documented problem 'thermal runaway' concerning aircraft lithium battery installation. User-submitted engineering problem (session input): "Why do aircraft lithium-battery installations suffer thermal events? (connector outage honesty)". [nasa_nts] 1 records retrieved live (2019-2019). Illustrative records: 'Statistical Characterization of 18650-Format Lithium-Ion Cell Thermal Runaway Energy Distributions'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Mechanical abuse poses a critical safety risk to Li-ion batteries by inducing internal short circuits that initiate thermal runaway. Remarkably, voltage recovery frequently emerges during thermal runa'; 'The pursuit of higher energy density in lithium-ion batteries has made high-nickel (Ni) layered oxides leading cathode candidates for next-generation electric vehicles. However, their poor thermal sta'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in thermal. 9-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 9-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Conduct flight-profile tests on instrumented battery packs with the PINN■based BMS enabled versus a conventional BMS; verify that no ther.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: temperature-rise mapping at max duty. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden; interface contact resistances dominate at small scale and must be measured, not assumed zero; steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k; one-dimensional gradient (lateral gradients neglected); TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.; TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.; TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.; Model input variables requiring sourced values before any numeric result: h ($W/(m^2.K)$), A (m^2), dT (K), L_i (m), k_i ($W/(m.K)$), A_i (m^2), k ($W/(m.K)$), A (m^2), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , $W/(m.K)$] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m^2] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Incorporation of Joule heating, reversible entropic heat, and radiative losses into a physics-informed neural network for real-time temperature prediction.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'thermal runaway' concerning aircraft lithium battery installation. User-submitted engineering problem (session input): "Why do aircraft lithium-battery installations suffer thermal events? (connector outage honesty)". [nasa_ntrs] 1 records retrieved live (2019-2019). Illustrative records: 'Statistical Characterization of 18650-Format Lithium-Ion Cell Thermal Runaway Energy Distributions'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Mechanical abuse poses a critical safety risk to Li-ion batteries by inducing internal short circuits that initiate thermal runaway. Remarkably, voltage recovery frequently emerges during thermal runa'; 'The pursuit of higher energy density in lithium-ion batteries has made high-nickel (Ni) layered oxides leading cathode candidates for next-generation electric vehicles. However, their poor thermal sta'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: thermal

Disciplines: heat transfer, thermal engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): heat source characterization; conduction path / spreader; reject interface; safety limit interlock — realizing: Embed the enhanced physics-informed neural network model into the aircraft lithium■battery management system to continuously predict cell surface temperature during flight and trigger protective actions (e.g., load reduction, active cooling) when predicted temperature approaches runaway thresholds.

Subsystems:

?: heat source characterization - ?

?: conduction path / spreader - ?

?: reject interface - ?

?: safety limit interlock - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Embed the enhanced physics-informed neural network model into the aircraft lithium■battery management system to continuously predict cell surface temperature during flight and trigger protective actions (e.g., load reduction, active cooling) when predicted temperature approaches runaway thresholds.

Key Physics: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Thermal management / heat transfer governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

TH-002: Convective heat transfer (Newton's law of cooling). Expression: $q = h * A * dT$. Applicability to this invention: APPLICABLE; holds under: steady state; h characterized for the actual flow geometry. Assumptions: h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden. Engaged variables: dT .

TH-003: Series thermal resistance. Expression: $R_{total} = \sum(L_i / (k_i * A_i))$. Applicability to this invention: APPLICABLE; holds under: one-dimensional steady conduction through layers. Assumptions: interface contact resistances dominate at small scale and must be measured, not assumed zero. Engaged variables: R_{total} .

TH-001: Fourier conduction. Expression: $q = -k * A * dT/dx$. Applicability to this invention: APPLICABLE; holds under: steady state, isotropic conduction. Assumptions: steady-state conduction (no transient term); isotropic, homogeneous medium conductivity k ; one-dimensional gradient (lateral gradients neglected). Engaged variables: k , dT/dx .

Assumptions:

- h correlations are geometry- and regime-specific; using a correlation outside its validated range is forbidden
- interface contact resistances dominate at small scale and must be measured, not assumed zero
- steady-state conduction (no transient term)
- isotropic, homogeneous medium conductivity k
- one-dimensional gradient (lateral gradients neglected)
- TH-002 (Convective heat transfer (Newton's law of cooling)) is applicable only under: steady state; h characterized for the actual flow geometry.
- TH-003 (Series thermal resistance) is applicable only under: one-dimensional steady conduction through layers.

- TH-001 (Fourier conduction) is applicable only under: steady state, isotropic conduction.
- Model input variables requiring sourced values before any numeric result: h (W/(m².K)), A (m²), dT (K), L_i (m), k_i (W/(m.K)), A_i (m²), k (W/(m.K)), A (m²), dT/dx (K/m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: q (W), R_{total} (K/W), q (W) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): heat generation at source [q , W] value_status=UNKNOWN; spreader conductivity [k_i , W/(m.K)] value_status=UNKNOWN; interface contact resistance [R_{total} , K/W] value_status=UNKNOWN; reject-path area [A , m²] value_status=UNKNOWN; allowable temperature rise [dT , K] value_status=UNKNOWN; ambient worst case [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'thermal runaway', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'hotspot exceeding material/safety limit', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'interface delamination raising resistance over life', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'coolant/flow path blockage (if active)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'thermal runaway coupling to electronics', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'condensation at cold surfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	prevent thermal runaway under flight conditions	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Incorporation of Joule heating, reversible entropic heat, and radiative losses	MODE_PHYSICS-informed neural network for r
DI-003	Expected effect target	Early detection of abnormal heating prevents temperature from reaching runaway conditions, thereby eliminating thermal	MODE_PHYSICS
DI-004	Domain design input pattern: distribution of expected system tokens ['system']		ENGINEERING_PROPOSED
DI-005	Domain design input pattern: ambient and operating conditions ['conditions']		ENGINEERING_PROPOSED
DI-006	Domain design input pattern: safety-critical application tokens ['safety', 'temperature']		ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance with standards (e.g. DO-160, DO-178, DO-254, etc.)	general electrical safety, UNKOWN (PERFORMANCE PROPOSED)	is never inferred
DI-008	Custodied observation: Development of a small engine parting article (32-5837) to take off in 1949	5608256335AEC2c11149f55b11cfe5908d34b	
DI-009	Custodied observation: From Present to Future Part 1 of 404 P0701 showing you may	5820b39b0cb5074P07017532df076abbcb94ff3478	
DI-010	Custodied observation: A Review of the Manufacture of Engines	40219267a30e5b1hastee58623af2050d03afef7756975b8211c00c878e98	
DI-011	Custodied observation: Advances in High Temperature Sensor Technology	6165e7MaterialSystem06e851g6019074e2ef1A9a519a249b91974afe93f01	
DI-012	Custodied observation: Low-Temperature Operation of Commercial Batteries	42222331Batteries_basing_5152nic0458g38C0FhF0271a3b8e3b04030634170d52	

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	heat source characterization realizing Embed the enhanced physics	CONFIDENTIAL	needed to lock the code into the aircraft lithium battery management system
DO-002	conduction path / spreader realizing Embed the enhanced physics	CONFIDENTIAL	needed to lock the code into the aircraft lithium battery management system
DO-003	reject interface realizing Embed the enhanced physics	CONFIDENTIAL	needed to lock the code into the aircraft lithium battery management system
DO-004	safety limit interlock realizing Embed the enhanced physics	CONFIDENTIAL	needed to lock the code into the aircraft lithium battery management system
DO-005	heat source characterization interfaces with conduction	CONFIDENTIAL	needed to lock the code into the aircraft lithium battery management system
DO-006	conduction path / spreader interfaces with reject interface	CONFIDENTIAL	needed to lock the code into the aircraft lithium battery management system
DO-007	reject interface interfaces with safety limit interlock	CONFIDENTIAL	needed to lock the code into the aircraft lithium battery management system
DO-008	Device boundaries: patient/fluid inlet and outlet terminal	CONFIDENTIAL	needed to lock the code into the aircraft lithium battery management system
DO-009	Flow/force/signal path geometry realizing the mechanism	CONFIDENTIAL	needed to lock the code into the aircraft lithium battery management system
DO-010	Operating range for heat generation at source within which the mechanism holds its stated behavior	UNKNOWN	needed to lock the code into the aircraft lithium battery management system
DO-011	Operating range for spreader conductivity within which the mechanism holds its stated behavior	UNKNOWN	needed to lock the code into the aircraft lithium battery management system
DO-012	Operating range for interface contact resistance within which the mechanism holds its stated behavior	UNKNOWN	needed to lock the code into the aircraft lithium battery management system
DO-013	Operating range for reject-path area within which the mechanism holds its stated behavior	UNKNOWN	needed to lock the code into the aircraft lithium battery management system
DO-014	Operating range for allowable temperature rise within which the mechanism holds its stated behavior	UNKNOWN	needed to lock the code into the aircraft lithium battery management system
DO-015	Operating range for ambient worst case within which the mechanism holds its stated behavior	UNKNOWN	needed to lock the code into the aircraft lithium battery management system
DO-016	Sensing -> decision -> actuation loop maintaining the proposed operating point	PROPOSED	needed to lock the code into the aircraft lithium battery management system
DO-017	Information pipeline: ingestion -> feature/representation	PROPOSED	needed to lock the code into the aircraft lithium battery management system
DO-018	Fixture enabling: temperature-rise mapping at max duty	UNKNOWN	needed to lock the code into the aircraft lithium battery management system
DO-019	Fixture enabling: interface resistance characterization	UNKNOWN	needed to lock the code into the aircraft lithium battery management system
DO-020	Fixture enabling: worst-case ambient soak test — rig control	UNKNOWN	needed to lock the code into the aircraft lithium battery management system

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: temperature-rise mapping at maximum	UNKNOWN	Unknown concept name; fixture specific instrumentation selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
heat generation at source	UNKNOWN (no sourced value)	W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
spreader conductivity	UNKNOWN (no sourced value)	W/(m.K)	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
interface contact resistance	UNKNOWN (no sourced value)	K/W	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
reject-path area	UNKNOWN (no sourced value)	m^2	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
allowable temperature rise	UNKNOWN (no sourced value)	K	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
ambient worst case	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
thermal runaway	Fails by thermal runaway: the safety shutdown is triggered, which is not a failure mode	Design failure NOT ESTIMATED (requires trigger) UNKNOWN (operating test conditions)	UNKNOWN (operating test conditions)	candidate failure mode (A) (m)
hotspot exceeding material safety limit	Fails by hotspot exceeding material safety limit; heat spreader spreading overheats and the hotspot modes (A) (s)	Design failure NOT ESTIMATED (requires trigger) UNKNOWN (operating test conditions)	UNKNOWN (operating test conditions)	candidate failure mode (A) (s)
interface delamination	Fails by interface delamination; mass electrical resistance over time; poor expansion with high power and high temperature	Design failure NOT ESTIMATED (requires trigger) UNKNOWN (operating test conditions)	UNKNOWN (operating test conditions)	candidate failure mode (A) (s)
coolant/flow path blockage (if active)	Fails by coolant/flow path blockage (if active); particulates in coolant block up active cooling channels; candidate failure modes (A) (t)	Design failure NOT ESTIMATED (requires trigger) UNKNOWN (operating test conditions)	UNKNOWN (operating test conditions)	candidate failure mode (A) (t)
thermal runaway coupling to electronics	Fails by thermal runaway coupling to electronics; making thermal analysis; raised leakage unknown, which is a candidate failure mode (C) (b)	Design failure NOT ESTIMATED (requires trigger) UNKNOWN (operating test conditions)	UNKNOWN (operating test conditions)	candidate failure mode (C) (b)
condensation at cold surfaces	Fails by condensation at cold surfaces; surfaces operating below dew point only; condensation; candidate failure and mode (C) (d)	Design failure NOT ESTIMATED (requires trigger) UNKNOWN (operating test conditions)	UNKNOWN (operating test conditions)	candidate failure and mode (C) (d)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
thermal runaway	Fails by thermal runaway: the device	design control	NOT ESTABLISHED
hotspot exceeding material safety limit	hotspot exceeding material/safety	under gear heavy gear	station on speed
interface delamination raising risk of system failure	interface delamination raising	selective real time cycle expansion	cycle life
coolant/flow path blockage (if active)	coolant/flow path blockage (if	active)	particulate to debris
thermal runaway coupling	Fails by thermal runaway coupling	to	electrical
condensation at cold surfaces	Fails by condensation at cold surfaces:	surface strength	the operating point

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Conduct flight-profile tests on instrument	Conduct flight-profile tests on instrument	Conduct flight-profile tests on instrument	Conduct flight-profile tests on instrument
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test	run killer experiment: falsification_test
VF-003	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty
VF-004	temperature-rise mapping at maximum	temperature-rise mapping at maximum	temperature-rise mapping at maximum	temperature-rise mapping at maximum
VF-005	thermal-resistance trend over acceleration	thermal-resistance trend over acceleration	thermal-resistance trend over acceleration	thermal-resistance trend over acceleration
VF-006	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty	temperature-rise mapping at max duty
VF-007	worst-case ambient soak with staged	worst-case ambient soak with staged	worst-case ambient soak with staged	worst-case ambient soak with staged

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	biocompatible thermal spreader (graphite/AlN)	graphite/AlN	electronics thermal practice	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
biocompatible thermal spreader	biocompatible thermal spreader	1	STANDARD	biocompatible thermal spreader	Not Established	STANDARD	Stability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- bonded spreader + encapsulation: domain registry manufacturing route - candidate only; no process qualification exists
- thermal interface material application control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter heat generation at source [q W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter spreader conductivity [k_i W/(m.K)]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter interface contact resistance [R_total K/W]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter reject-path area [A m^2]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter allowable temperature rise [dT K]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter ambient worst case [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}

- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	maintaining the proposed physical verification	conducting the proposed physical verification	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance	NOT ESTABLISHED (to be quoted)	constraint-compliance	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Conduct flight-profile	NOT ESTABLISHED (to be quoted)	VF-001 packaging and assembly	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment:	NOT ESTABLISHED (to be quoted)	VF-002 article	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	temperature-rise mapping	NOT ESTABLISHED (to be quoted)	VF-003 verification	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	temperature-rise mapping	NOT ESTABLISHED (to be quoted)	VF-004 verification	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	thermal-resistance trend	NOT ESTABLISHED (to be quoted)	VF-005 measurement	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	temperature-rise mapping	NOT ESTABLISHED (to be quoted)	VF-006 verification	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	worst-case ambient	NOT ESTABLISHED (to be quoted)	VF-007 verification	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 9-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.